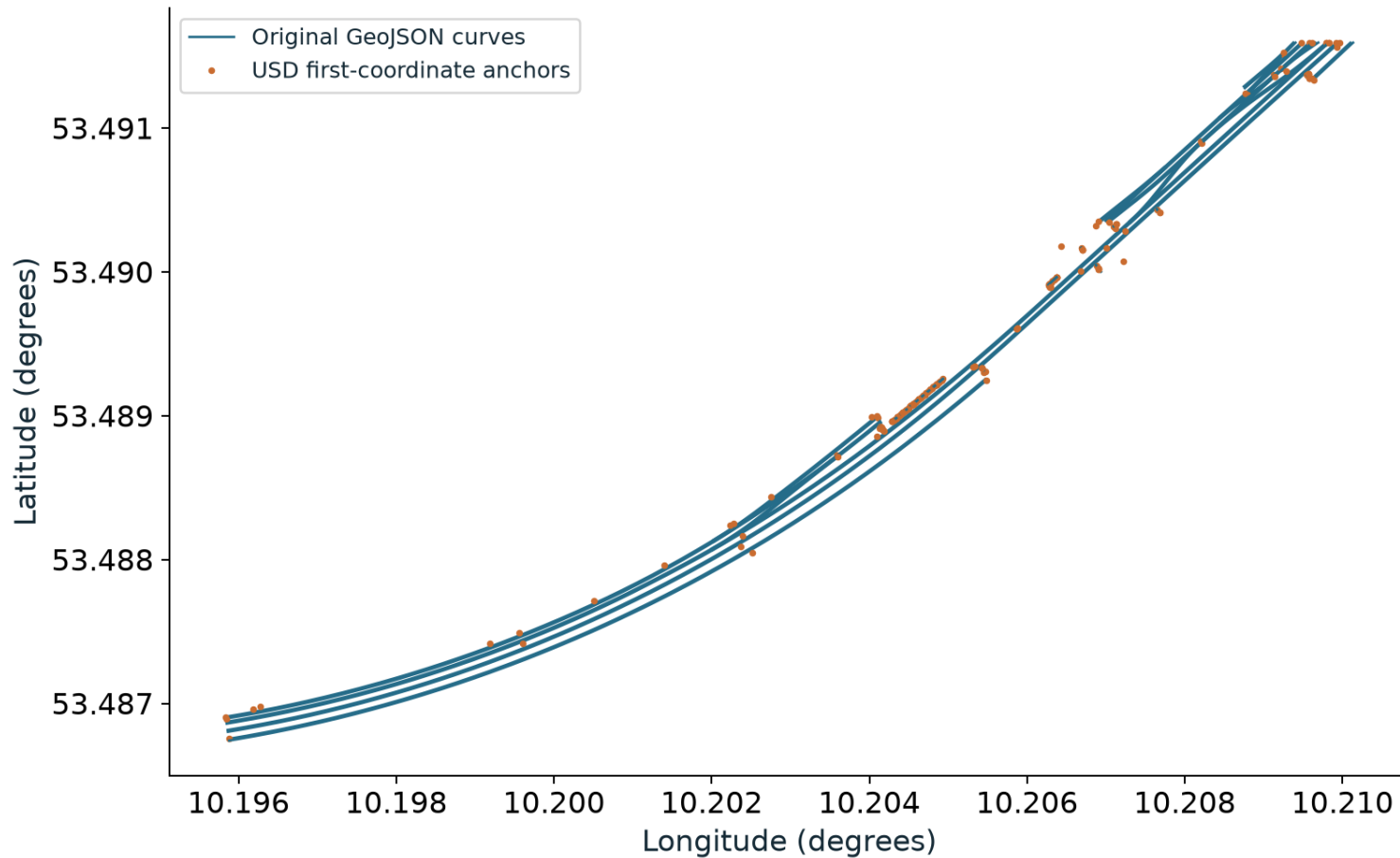


Geospatial scenes with shared placement

A railway, a site asset and a scalar field should retain their source coordinates while participating in one scene.

Rendering, world-position queries, bounds and physics need the same resolved placement.

Railway source and scene identity



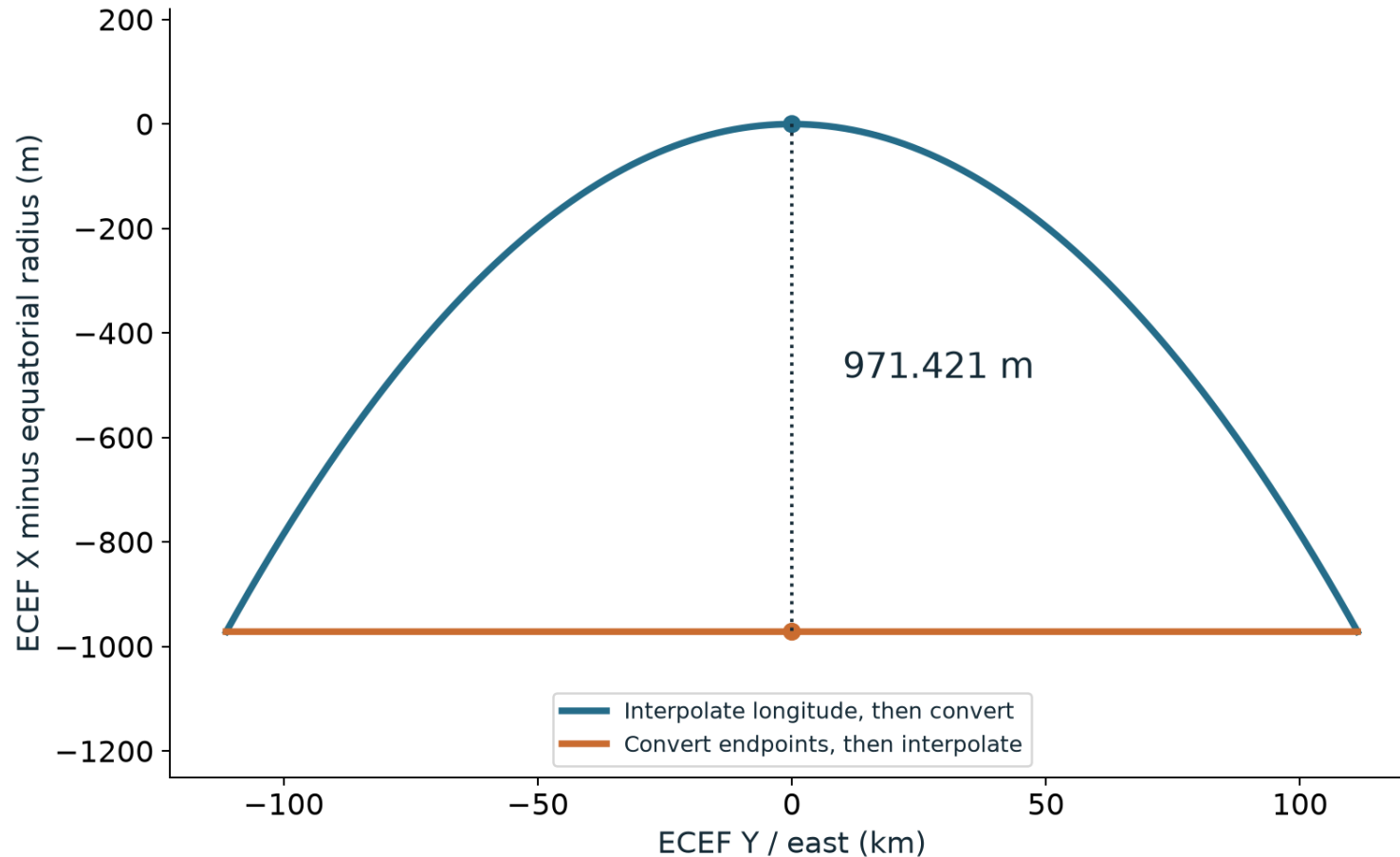
Original source curves and matching USD anchors; this is a data plot, not a runtime render.

1,473 source features match USD object identities and first-coordinate anchors.

186 railway curves retain 3,526 vertices by count; interior placement is still unchecked.

These checks establish what survived conversion; they do not establish resolved rails-on-map placement.

Native interpolation preserves the authored path



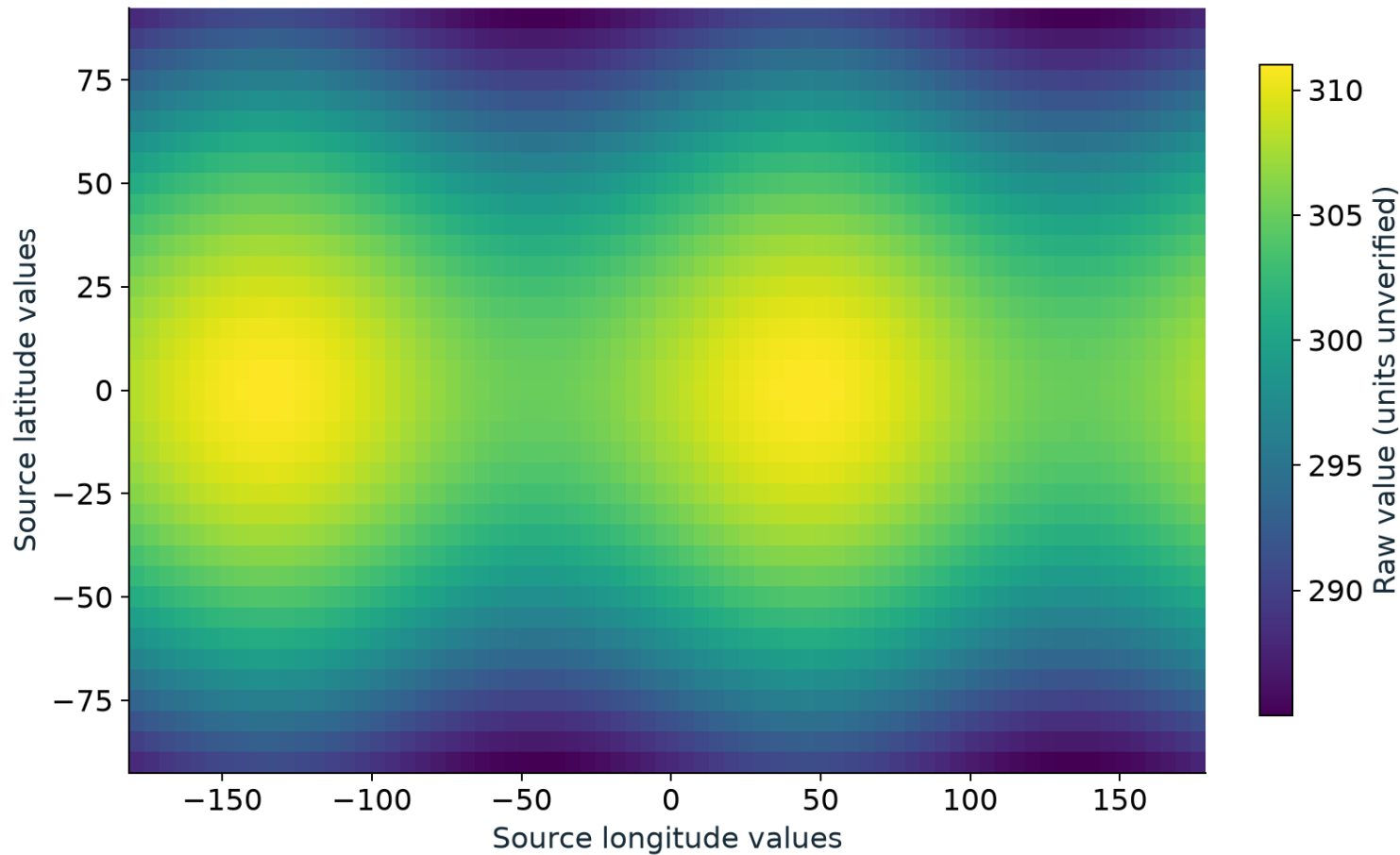
R20, Positions between recorded moments: interpolate recorded coordinates in their CRS, then convert.

Interpolating converted endpoints puts the equatorial midpoint 971.421 m inside the ellipsoid.

This is an analytic component example; the scene position carrier and its sampling rules remain open.

Equatorial samples at minus and plus one degree; vertical scale is exaggerated and axes use different units.

Scalar data and a removable visualization



2,664 scalar samples on a 37 by 72 grid exercise coordinate reporting; a three-sample overlay is removable.

The stock-USD overlay leaves source values intact. Geospatial placement of that overlay is still pending.

Forecast origin, timestamp, CRS metadata and value units are unverified; the filename is not provenance.

Raw values from the retained scalar-field fixture, without an inferred physical unit.

Proposed runtime contract

Composed declarations establish CRS scope; source coordinates and asset conventions remain authored data.

Resolution evaluates native positions and offsets into one selected output CRS without modifying the stage.

Queries and consumers share the result; failures and operation provenance remain explicit.

Two runtime targets

Non-Hydra: a renderer-independent scene resolver for placements, coordinate queries and bounds. Not built.

Hydra: a scene-index adapter consuming the same resolved placement and propagating changes. Not built.

Consumer consistency is distinct from R29, Implementable from the text alone, which requires different transformation engines.

Next complete demonstration

Import the railway and a site asset, resolve them into one project CRS, and verify placement against independent controls.

Relocate an asset, scrub time and compose another layer; compare non-Hydra queries with Hydra results after each change.

State placement error and valid extent, preserve source data, and fail visibly when a required operation cannot run.

Review requests

Review the proposed runtime prose against the functional requirements and close the carrier, placement and scope decisions.

Supply source CRS/epoch/vertical metadata, expected control points and a distance-and-extent acceptance criterion.

Use the next railway/site demonstration to review both consumer paths and separate engine agreement from correctness.